CV Intern Task: Egg Segmentation and Tracking

Anant | Nosh Robotics assignment | Repository:
https://github.com/anantsharma67/egg-segmentation-tracking

1. Data and split (leakage control)

The Roboflow export has 110 images (640x640) in classification-folder format, i.e. **no masks or boxes**. Roboflow's own train/valid/test folders were ignored and all images pooled. A perceptual-hash (pHash 16x16) check found one near-duplicate pair and no augmented copies, but many photos share a scene (pink bowl, steel pot, keyboard, rug). A random split would put the same background in both parts, so **make_split.py** groups images by (a) near-duplicate hash and (b) background colour (KMeans, k=24, on an HSV histogram) and assigns whole groups to a split. Result: 24 groups, **dev = 89 images, eval = 21 images (4.2 : 1)**.

2. Task 1: pretrained segmentation (pseudo ground truth)

Because there are no annotations, Grounding DINO (tiny, prompt "an egg.") proposes boxes and SAM ViT-B segments each box (task1_sam.py). It found eggs in 109/110 images (88 with one egg, 12 with two, 9 with three). I reviewed all six contact sheets by eye. Problems found and handled (fix_masks.py, logged in fixes.json): (i) whole-water-surface masks in the steel-pot photos (#124, #135, #145) and container/plate masks (#31, #40) were dropped by an area rule (>40% of image, or >30% with low detector score); (ii) white eggs on pale backgrounds missed by the detector (#105, #134) were re-prompted with manual boxes; (iii) egg-tray photos where no single egg is clearly visible (#116, #42) were excluded. Eggs in greenery (#10-#13) have holes where leaves occlude the egg; kept, noted as a SAM limitation. After cleanup, **108 images** remain with **137 egg masks** (90 images with one egg, 7 with two, 11 with three): 5 instances dropped, 4 eggs added with manual boxes (#105, #134, #124, #145; checked visually), 2 images excluded.

3. Task 2: classical segmentation and experiment log

Final algorithm (**t10** in classical.py): (1) generate ~11 candidate foreground masks by Otsu thresholding L, a, b, S, V channels in both polarities plus a Lab-distance-from-border-colour map; (2) clean each with open/close and hole filling; (3) score every connected component by ellipse fit x solidity x aspect plausibility, with a penalty per image border touched; keep components scoring ≥ 0.6 greedily by score; (4) refine with GrabCut at half resolution, seeded by those components; (5) if nothing passed, fall back to a central-rectangle GrabCut at 0.4x scale, and as a last resort the largest GrabCut blob. Metrics below are on the **dev split only** (87 images that have a ground-truth mask; 2 dev images were excluded); mIoU is the mean per-image foreground IoU against the Task 1 masks.

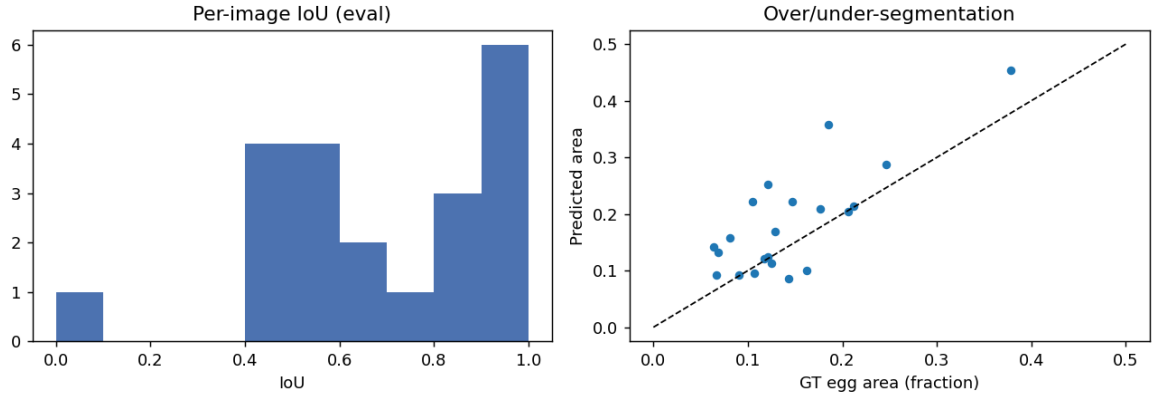
Trial	Change and why	mIoU	Median	Count acc.	Decision
t1	Grayscale Gaussian blur + Otsu. Simplest baseline: eggs are bright.	0.369	0.261	0.46	Baseline
t2	Lab distance from median border colour + Otsu. Assumes eggs differ from the background in colour.	0.329	0.275	0.20	Worse; many fragments
t3	t2 + keep only blobs with solidity ≥ 0.85 and area ≤ 50%. Remove non-egg shapes.	0.346	0.222	0.46	IoU about flat; shape filter kept
t4	t3 as seeds for full-resolution GrabCut. Sharpen boundaries.	0.364	0.185	0.46	Dropped: no gain, seeds often empty or wrong
t5	GrabCut from a central rectangle (first 29 dev images only, about 7 s/img). Bypass thresholds.	0.340	0.288	0.69	Too slow alone; reused as a low-res fallback
t6	11 candidate masks + ellipse-fit scoring (threshold sweep 0.5/0.6/0.7 gave 0.421/0.422/0.420).	0.422	0.423	0.48	Kept, thr = 0.6
t8	t6 components as seeds for half-res GrabCut. Diagnosis showed 28/85 empty predictions.	0.453	0.473	0.49	Kept
t9	t8 + central-rectangle GrabCut fallback (0.4x) when t6 finds nothing.	0.590	0.697	0.72	Kept, big gain

Trial	Change and why	mIoU	Median	Count acc.	Decision
t10	t9 + largest GrabCut blob when every filter rejects. Final.	**0.592**	**0.696**	**0.76**	**Final**

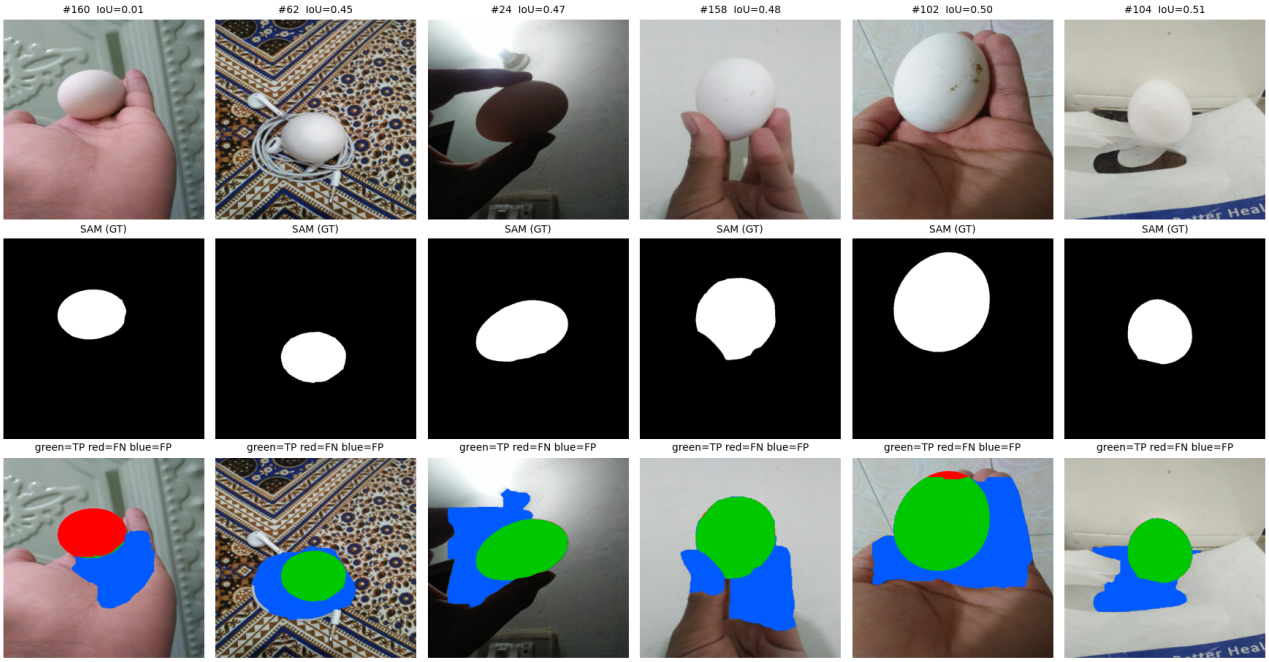
Trial numbering follows the function names in classical.py (there is no t7). The biggest lesson from the log: failures were dominated by *empty* predictions caused by my own shape filters, not by poor thresholds; fixing that (t8 to t9) mattered more than any boundary refinement.

4. Task 3: evaluation on the held-out eval split

Task 1 masks are treated as ground truth; the final classical method was run once on the 21 eval images (no tuning on eval). **mIoU 0.687, median IoU 0.687, Dice 0.784, precision 0.742, recall 0.891, boundary F1 (3 px tolerance) 0.567, instance-count accuracy 0.86.** 9 images have IoU ≥ 0.8, 7 are 0.5 to 0.8, and 5 are below 0.5. Recall (0.89) is much higher than precision (0.74): the method nearly always finds the egg but over-segments.



Qualitative analysis. Worst six eval images (red = missed egg pixels, blue = false positive, green = correct):



What went wrong. (a) **Leakage into skin and neighbouring surfaces** (#24, #158, #102, #104): the GrabCut fallback has no shape constraint, so for white eggs held in a hand or on a white bag it absorbs skin or paper of similar colour; the egg itself is segmented well (large green region) but the blue regions halve the IoU. (b) **Colour-similar egg and background** (#160): a pale pink egg on a skin-coloured hand was missed completely while the hand was segmented (IoU 0.01). (c) **Halo around thin occluders** (#62): the earphone cable around the egg is partly included. **Improvements I would try:** constrain the final mask to an ellipse fitted to the high-confidence core (distance-transform peak) so leaks are clipped; add a skin-colour prior in YCrCb to

suppress hands; use a texture/edge term in the energy for white-on-white eggs; and tune the fallback only after the leakage fix.

5. Task 4: video tracking

Video: "Raw and Boiled Egg Spinning" (Mocomi, 60 s, 1280x720, 24 fps). Output video (public Drive link): **https://drive.google.com/file/d/1rNWnFlvnG8v98lojL3CRERxVpdzDT91U/view?usp=drive_link**. The general classical segmenter (t9) fails on this clip: the eggs are small and the dark table and checked shirt dominate, so GrabCut segmented the table (probe frames showed red masks over the table and shirt). I therefore added a clip-specific segmenter (**segment_video_frame** in track_video.py), explicitly allowed by the task: (1) skip frames whose lower region is light (title and text cards contain no eggs); (2) in the table region keep bright, low-saturation pixels (V > 165, S < 75); (3) split touching eggs per connected component with a distance-transform watershed, and keep blobs that are ellipse-like (ellipse score ≥ 0.5, aspect ≥ 0.4, area 0.4% to 10% of the frame); (4) grow each core out to the shaded egg edge with a hysteresis step (V > 95, S < 95, max 14 px). Tracking (final version): Hungarian matching on (1 - IoU) with a 140 px centroid gate for visible tracks, lost tracks re-matched by egg size, and tracks kept for 80 processed frames so a hand hiding an egg does not break its ID (see the first attempt below). Frames were processed at stride 2.

Result. The clip is a single continuous shot with two eggs that a person moves across the table. The tracker creates exactly **2 track IDs** (egg 1 and egg 2); both are followed from their first appearance to the end of the egg section (frames 222 to 1000 of the 1440 in the clip), and never more than two objects are tracked at once. In 263 of the 380 processed frames that contain eggs both eggs are tracked together; in the other 117 only one egg is visible or segmented, mostly while the hand covers the other. **Limitations:** the two eggs look identical, so when both are lifted and put down in new places, the identity (which one is the raw egg) can be swapped; I could not verify this beyond watching the output. The segmentation thresholds are tuned to this clip (white eggs on a dark table); in a new scene they would need re-tuning or the general t9 segmenter.

First attempt: 4 egg IDs instead of 2, and the fix. My first tracker produced **4 track IDs for the 2 eggs** in this clip. Per-frame detections never exceeded two eggs, so the segmenter was fine; the extra IDs were a tracking error. The tracker only remembered a lost egg for 20 processed frames and only matched detections within 140 px of the last position. When the man's hand covered an egg and carried it across the table, the egg reappeared about 85 frames later and over 300 px away, so it was issued a brand-new ID (this happened twice, giving IDs 3 and 4). I first misread this as a cut to a second take, but the clip is one continuous shot with two eggs. **The fix that brought it to exactly 2 IDs:** (1) a track that has been lost for more than 2 frames is re-matched by *egg size* (area ratio) instead of position, since the eggs keep the same apparent size while moving; and (2) tracks are kept alive for 80 processed frames instead of 20 so that an egg hidden under a hand is still available to be re-matched.

6. AI-tool disclosure

I used Claude (Anthropic) to help write the scripts, to review SAM output, and to draft this report; I ran Task 1 on Colab myself, reviewed the masks, and can explain every stage of the pipeline above. Grounding DINO and SAM are third-party pretrained models used as published.

7. Reproduction

See README.md: install requirements, run make_split.py, task1_sam.py (GPU), fix_masks.py, then run_trial.py / task3_eval.py for the classical method and track_video.py for Task 4.